Business Question and Visualization Report

Global Olympic Sports Performance Analysis (1992–2026)

Date: 24 September 2026	**Team ID:** PNT2026TMID092418
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 5 Marks

Executive Summary & Analytical Framework

This report presents 8 core business questions designed to evaluate Olympic athletic performance, resource allocation, investment efficiency, and host nation advantage using Snowflake analytical views and Tableau / Streamlit visual outputs.

BQ-1: Which sports categories deliver the highest average medals per athlete across Olympic Games?

SQL Query (Snowflake):

```
SELECT sport_category,
COUNT(medal_id) AS total_medals,
COUNT(DISTINCT athlete_id) AS total_athletes,
ROUND(COUNT(medal_id)::FLOAT / NULLIF(COUNT(DISTINCT athlete_id), 0), 3) AS avg_medals_per_athlete
FROM OLYMPIC_PERFORMANCE_DB.MART.FACT_MEDAL_ACHIEVEMENTS
GROUP BY sport_category
ORDER BY avg_medals_per_athlete DESC
LIMIT 10;
```

Visualization Type: Horizontal Bar Chart (Tableau / Streamlit)
Visual Output Summary: Rowing ranks highest at 6.916 average medals per athlete, closely followed by Shooting at 6.731. Demonstrates that technical, multi-event, or small-roster team sports provide the highest per-athlete medal returns for NOC investment.

BQ-2: Which host nations achieved the highest total medal tallies during their home Olympic Games?

SQL Query (Snowflake):

```
SELECT host_country,
SUM(CASE WHEN delegation_country = host_country THEN 1 ELSE 0 END) AS host_nation_medals
FROM OLYMPIC_PERFORMANCE_DB.MART.FACT_MEDAL_ACHIEVEMENTS
GROUP BY host_country
ORDER BY host_nation_medals DESC
LIMIT 5;
```

Visualization Type: Comparative Multi-Bar Chart
Visual Output Summary: Japan leads home nation dominance with 105 host medals, followed by the United States with 100 medals and China with 93 medals, highlighting significant home advantage effects.

BQ-3: How do top-performing countries compare in cumulative total medal count from 1992 to 2026?

SQL Query (Snowflake):

```
SELECT country_name, COUNT(medal_id) AS total_medals
FROM OLYMPIC_PERFORMANCE_DB.MART.FACT_MEDAL_ACHIEVEMENTS
GROUP BY country_name
ORDER BY total_medals DESC
LIMIT 10;
```

Visualization Type: Ranked Vertical Column Chart
Visual Output Summary: The United States leads globally with 735 total medals, followed by China with 696 medals, establishing the global competitive benchmark across modern Olympic history.

BQ-4: What is the relationship between national sports capital investment and actual medal counts?

SQL Query (Snowflake):

```
SELECT country_name, region, sports_investment_usd, total_medals,
ROUND((total_medals / NULLIF(sports_investment_usd, 0)) * 1000000, 2) AS efficiency_ratio
FROM OLYMPIC_PERFORMANCE_DB.MART.VW_INVESTMENT_EFFICIENCY
ORDER BY efficiency_ratio DESC;
```

Visualization Type: Global Sports Investment vs. Medal Performance Bubble Chart
Visual Output Summary: Highlights overperforming nations like Slovakia, Georgia, and Ethiopia, whose bubble positions demonstrate disproportionately high medal returns relative to low financial budgets.

BQ-5: How does medal efficiency percentage vary across geographic regions?

SQL Query (Snowflake):

```
SELECT region,
ROUND(AVG(efficiency_percentage), 2) AS avg_regional_efficiency_pct
FROM OLYMPIC_PERFORMANCE_DB.MART.VW_INVESTMENT_EFFICIENCY
GROUP BY region
ORDER BY avg_regional_efficiency_pct DESC;
```

Visualization Type: Regional Efficiency Pie & Donut Chart
Visual Output Summary: South America leads regional efficiency at 49.64%, followed by Europe/Asia at 45.98%, showing that South American athletic programs extract maximum performance per dollar invested.

BQ-6: What is the historical trend of total medals awarded per Olympic Games cycle from 1992 to 2026?

SQL Query (Snowflake):

```
SELECT olympic_year, COUNT(medal_id) AS total_medals_awarded
FROM OLYMPIC_PERFORMANCE_DB.MART.FACT_MEDAL_ACHIEVEMENTS
GROUP BY olympic_year
ORDER BY olympic_year ASC;
```

Visualization Type: Time-Series Line Chart
Visual Output Summary: Traces medal totals swinging from a low of 305 in Sydney 2000 to a peak of 363 in Tokyo 2020, and recording 334 in Milano-Cortina/Dakar 2026.

BQ-7: How are medals distributed between Summer and Winter Olympic Games by region?

SQL Query (Snowflake):

```
SELECT region, game_season, COUNT(medal_id) AS medals_won
FROM OLYMPIC_PERFORMANCE_DB.MART.FACT_MEDAL_ACHIEVEMENTS
GROUP BY region, game_season
ORDER BY region, game_season;
```

Visualization Type: Stacked Segmented Bar Chart
Visual Output Summary: Demonstrates that Europe and Asia dominate Winter Games medals, while North America and South America show strong concentration in Summer Games disciplines.

BQ-8: Which gender demographics account for the growth in athlete participation and medal wins?

SQL Query (Snowflake):

```
SELECT olympic_year, gender, COUNT(DISTINCT athlete_id) AS total_athletes, COUNT(medal_id) AS total_medals
FROM OLYMPIC_PERFORMANCE_DB.MART.FACT_MEDAL_ACHIEVEMENTS
GROUP BY olympic_year, gender
ORDER BY olympic_year ASC, gender;
```

Visualization Type: Dual-Axis Area & Line Chart
Visual Output Summary: Illustrates a steady increase in female athlete participation and medal share from 1992 to 2026, reaching near 50-50 gender parity in recent Games.